

PART 1 / EXPECTATION AND VARIANCE

From affine rules to a variance bound

The setting. The question assumes independent losses $Z_1, \dots, Z_n \in [0, 1]$ with common mean $\mu = \mathbb{E}[Z_i]$. Define

$$\widehat{\mu}_n = \frac{1}{n} \sum_{i=1}^n Z_i, \quad D_n = |\widehat{\mu}_n - \mu|.$$

We use n for sample size and i for the observation index. The informal expression $|\mu_i - \mu|$ from the discussion is interpreted here as the random estimation error D_n . If both means were fixed population parameters, their difference would be constant and its variance would be zero.

1.1 What changes when $Y = aX + b$?

For constants a, b and a random variable X with finite variance,

$$\mathbb{E}[Y] = a\mathbb{E}[X] + b, \quad \text{Var}(Y) = a^2 \text{Var}(X).$$

The variance identity follows by centering:

$$Y - \mathbb{E}[Y] = a(X - \mathbb{E}[X]), \quad \mathbb{E}[(Y - \mathbb{E}[Y])^2] = a^2 \mathbb{E}[(X - \mathbb{E}[X])^2].$$

Adding a constant changes the mean but not the variance. Scaling by a multiplies the variance by a^2 . Absolute value is not an affine transformation, so this rule alone does not determine $\text{Var}(D_n)$.

1.2 Why does a variable in $[0, 1]$ have variance at most $1/4$?

Let $0 \leq W \leq 1$ almost surely. Squaring and then taking expectations gives

$$W^2 \leq W \implies \mathbb{E}[W^2] \leq \mathbb{E}[W].$$

Substituting this upper bound into the variance formula gives

$$\begin{aligned} \text{Var}(W) &= \mathbb{E}[W^2] - (\mathbb{E}[W])^2 \\ &\leq \mathbb{E}[W] - (\mathbb{E}[W])^2 \\ &= m - m^2 \quad \text{where } m = \mathbb{E}[W] \\ &= \frac{1}{4} - \left(m - \frac{1}{2}\right)^2 \leq \frac{1}{4}. \end{aligned}$$

This completes the “squaring makes it smaller” argument: the subtraction of m^2 is essential. The general result is $\leq 1/4$, with equality for a variable taking 0 and 1 with equal probability.

Since $Z_i \in [0, 1]$ and $D_n \in [0, 1]$, the bound applies to both. More precisely, if $0 < \mu < 1$, then $D_n \leq M = \max(\mu, 1 - \mu) < 1$, so scaling gives $\text{Var}(D_n) \leq M^2/4 < 1/4$. At $\mu = 0$ or 1 , all losses equal μ almost surely and $D_n = 0$.

Independence yields the useful sample-size bound

$$\mathbb{E}[\widehat{\mu}_n] = \mu, \quad v_n := \text{Var}(\widehat{\mu}_n) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(Z_i) \leq \frac{1}{4n}.$$

Also, $\mathbb{E}[D_n^2] = v_n$. This is the variance of the sample mean, not the variance of its absolute error:

$$\text{Var}(D_n) = v_n - (\mathbb{E}[D_n])^2 \leq v_n.$$

PART 2 / BOUNDS, POWERS, AND LIMITS

What convexity and concavity tell us

2.1 A computable interval from a known second moment

Start with a generic random variable $0 \leq X \leq 1$. Set

$$m = \mathbb{E}[X], \quad q = \mathbb{E}[X^2].$$

The pointwise comparison $X^2 \leq X$ gives $q \leq m$. For the other direction, Jensen's inequality for the concave function $u \mapsto \sqrt{u}$ gives

$$m = \mathbb{E}[\sqrt{X^2}] \leq \sqrt{\mathbb{E}[X^2]} = \sqrt{q}.$$

Equivalently, Jensen for the convex function $x \mapsto x^2$ gives $m^2 \leq q$. Thus

$$q \leq \mathbb{E}[X] \leq \sqrt{q}.$$

These are bounds on the **expectation**, not deterministic bounds on every realization of X . The lower bound uses the range $[0, 1]$; the upper bound uses Jensen.

For **Jensen's inequality**, the directions are: $\phi(\mathbb{E}[X]) \leq \mathbb{E}[\phi(X)]$ for convex ϕ , and the reverse inequality for concave ϕ .

2.2 Can these assumptions determine a single value?

Generally, no. For any $0 < q < 1$, both of the following have second moment q :

Distribution of X	$\mathbb{E}[X^2]$	$\mathbb{E}[X]$
$X = \sqrt{q}$ with probability one	q	$\sqrt{q}$
$X = 1$ with probability q , otherwise 0	q	q

So the interval cannot be universally narrowed using only $X \in [0, 1]$ and its second moment. For example, $q = 1/4$ permits expectations $1/4$ and $1/2$. A unique value follows at $q = 0$ or $q = 1$, or from additional distributional information.

2.3 Apply the result to the estimation error

With $X = D_n$ and $q = v_n$, we obtain

$$v_n \leq \mathbb{E}[D_n] \leq \sqrt{v_n} \leq \frac{1}{2\sqrt{n}}.$$

Here v_n is the actual variance. Since the question supplies only an *upper bound* on it, we cannot replace v_n in the lower bound by $1/(4n)$. Deterministic losses $Z_i = \mu$ give $\mathbb{E}[D_n] = v_n = 0$ for every n .

The generic sharpness example above is not an assertion that every such X is an estimation error for a fixed μ . Even within the question's model, however, the second moment need not identify the absolute-error expectation. For $n = 1$ and $\mu = 1/2$, equal probabilities at $Z_1 = 0.3, 0.7$ give $v_1 = 0.04$ and $\mathbb{E}[D_1] = 0.2$. Probabilities $0.08, 0.84, 0.08$ at $Z_1 = 0, 0.5, 1$ give the same mean and variance, but $\mathbb{E}[D_1] = 0.08$.

PART 2 / CONTINUED

Does letting the power approach one help?

2.4 The proposed power sandwich is valid

For $0 \leq X \leq 1$ and $t > 1$,

$$X^t \leq X \leq X^{1/t}, \quad \mathbb{E}[X^t] \leq m \leq \mathbb{E}[X^{1/t}].$$

As $t \downarrow 1$, both transformed variables converge to X . All are bounded by 1, so the dominated convergence theorem justifies taking limits inside expectations:

$$\lim_{t \downarrow 1} \mathbb{E}[X^t] = m = \lim_{t \downarrow 1} \mathbb{E}[X^{1/t}].$$

Thus the two exact moment bounds really do meet. But their common limit is the unknown m itself. Without knowing or independently evaluating those moments, this identifies a relationship rather than a numerical value.

2.5 Jensen bounds the powers, but does not remove the unknown

Convexity of x^t and concavity of $x^{1/t}$ give the full chain

$$m^t \leq \mathbb{E}[X^t] \leq m \leq \mathbb{E}[X^{1/t}] \leq m^{1/t}.$$

Every outer endpoint still contains m . To use *known* information, suppose again that only $q = \mathbb{E}[X^2]$ is available. For $1 < t \leq 2$, Jensen twice gives

$$m \leq (\mathbb{E}[X^t])^{1/t}, \quad \mathbb{E}[X^t] = \mathbb{E}[(X^2)^{t/2}] \leq q^{t/2}.$$

The first step uses convexity of x^t ; the second uses concavity of $u^{t/2}$ (linearity at $t = 2$). Combining them,

$$m \leq (q^{t/2})^{1/t} = \sqrt{q}.$$

The computable upper bound is unchanged as $t \downarrow 1$. The lower bound $q \leq m$ is unchanged too. **A tunable power does not create missing moment information.**

2.6 Two different limits, with different conclusions

For fixed n , the power limit $t \downarrow 1$ approaches $\mathbb{E}[D_n]$, an unknown finite-sample quantity. In contrast, the sample-size limit has explicit endpoints:

$$0 \leq \mathbb{E}[D_n] \leq \frac{1}{2\sqrt{n}} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

The squeeze theorem therefore proves $\mathbb{E}[D_n] \rightarrow 0$, called convergence in mean absolute error, or L^1 convergence of $\hat{\mu}_n$ to μ . Markov then gives, for each fixed $\varepsilon > 0$,

$$\mathbb{P}(D_n \geq \varepsilon) \leq \frac{\mathbb{E}[D_n]}{\varepsilon} \leq \frac{1}{2\varepsilon\sqrt{n}} \rightarrow 0.$$

Verdict on the idea. Convexity and concavity provide a valid interval. Exact power moments squeeze to the desired expectation as $t \downarrow 1$, but do not calculate it from the stated information. Letting $n \rightarrow \infty$ does prove a known limit, namely zero. Neither argument alone gives an exact finite-sample value or an exact convergence rate.

PART 2 / APPLICATION TO QUESTION 2

From moments to a defensible sample size

The laboratory requires $\mathbb{P}(D_n \geq 0.1) \leq 0.05$. Markov states that, for $Y \geq 0$ and $a > 0$, $\mathbb{P}(Y \geq a) \leq \mathbb{E}[Y]/a$. The transformation of the error determines which moment enters the bound.

2.7 Method M: absolute error, with Jensen

Choose $Y = D_n$ and $a = \varepsilon$. The bound from page 2 gives

$$\mathbb{P}(D_n \geq \varepsilon) \leq \frac{\mathbb{E}[D_n]}{\varepsilon} \leq \frac{1}{2\varepsilon\sqrt{n}}.$$

At $\varepsilon = 0.1$, requiring the displayed bound to be at most 0.05 gives

$$\frac{5}{\sqrt{n}} \leq 0.05 \iff \sqrt{n} \geq 100 \iff \boxed{n \geq 10,000}.$$

Direct Markov is therefore useful when supplemented by Jensen and the variance calculation. The crude bound $\mathbb{E}[D_n] \leq 1$ alone would instead give a trivial probability bound.

2.8 Squared-error Markov is Method C: Chebyshev

Choose $Y = D_n^2$ and $a = \varepsilon^2$. Since $\mathbb{E}[D_n^2] = v_n$,

$$\mathbb{P}(D_n \geq \varepsilon) = \mathbb{P}(D_n^2 \geq \varepsilon^2) \leq \frac{v_n}{\varepsilon^2} \leq \frac{1}{4n\varepsilon^2}.$$

This is precisely Chebyshev's inequality applied to $\hat{\mu}_n$. Consequently,

$$\frac{25}{n} \leq 0.05 \iff \boxed{n \geq 500}.$$

Markov does not force the choice $Y = D_n$. Applying it to the square already recovers the stronger Chebyshev calculation; no power limit is required.

2.9 Method H: Hoeffding

Independence and $Z_i \in [0, 1]$ give the two-sided bound

$$\mathbb{P}(D_n \geq \varepsilon) \leq 2 \exp(-2n\varepsilon^2).$$

Thus $2e^{-0.02n} \leq 0.05$ is equivalent to

$$n \geq \frac{\log 40}{0.02} \approx 184.444, \quad \boxed{n \geq 185} \text{ for integer } n.$$

Calculation	Bound at $\varepsilon = 0.1$	Sufficient n
Absolute-error Markov + Jensen	$\min\{1, 5/\sqrt{n}\}$	10,000
Squared-error Markov / Chebyshev	$\min\{1, 25/n\}$	500
Hoeffding	$\min\{1, 2e^{-0.02n}\}$	185

Recommendation. Under the question's assumptions, $n = 185$ is sufficient, since $2e^{-3.7} \approx 0.04945 < 0.05$. It is the smallest integer certified by this Hoeffding bound, not necessarily the true minimum required by the loss distribution. Likewise, 10,000 is a threshold for the displayed Markov–Jensen certificate, not an infimum of the expected error.

PART 3 / TWO RELATED CONCEPTS

Suprema, maxima, and dependence

3.1 Supremum versus maximum

A maximum is a largest value *attained* in a set. A supremum is its least upper bound and need not belong to the set. For example,

$$\sup(0, 1) = 1 \quad \text{but no maximum exists,} \quad \max[0, 1] = \sup[0, 1] = 1.$$

Similarly, an infimum is the greatest lower bound; a minimum must be attained. Every nonempty bounded subset of the real numbers has a supremum and an infimum, but may lack a maximum or minimum.

The power idea gives a concrete connection. For any $0 \leq X \leq 1$,

$$\sup_{t>1} \mathbb{E}[X^t] = \mathbb{E}[X] = \inf_{t>1} \mathbb{E}[X^{1/t}].$$

If $\mathbb{P}(0 < X < 1) > 0$, each lower moment is strictly smaller than $\mathbb{E}[X]$ and each upper moment is strictly larger. Neither extremum is attained for $t > 1$; they are approached as $t \downarrow 1$. If X takes only 0 and 1, all these moments coincide and are attained.

3.2 Where do the cross terms come from?

Let $Z_1, \dots, Z_n$ have finite variances and let a_i be real constants. We derive the formula without assuming independence.

Step 1: subtract the mean. Write $U_i = Z_i - \mathbb{E}[Z_i]$ and $S = \sum_{i=1}^n a_i Z_i$. By linearity of expectation,

$$S - \mathbb{E}[S] = \sum_{i=1}^n a_i (Z_i - \mathbb{E}[Z_i]) = \sum_{i=1}^n a_i U_i.$$

Step 2: expand the square. For two terms, $(a_1 U_1 + a_2 U_2)^2 = a_1^2 U_1^2 + 2a_1 a_2 U_1 U_2 + a_2^2 U_2^2$. With n terms, the same expansion is

$$\left(\sum_{i=1}^n a_i U_i \right)^2 = \sum_{i=1}^n a_i^2 U_i^2 + 2 \sum_{i<j} a_i a_j U_i U_j.$$

The factor 2 appears because each distinct pair occurs twice: as (i, j) and as (j, i) . The notation $i < j$ counts each pair just once.

Step 3: take expectations. Using the definition of variance and linearity,

$$\begin{aligned} \text{Var}(S) &= \mathbb{E}[(S - \mathbb{E}[S])^2] \\ &= \mathbb{E} \left[\sum_i a_i^2 U_i^2 + 2 \sum_{i<j} a_i a_j U_i U_j \right] \\ &= \sum_i a_i^2 \mathbb{E}[U_i^2] + 2 \sum_{i<j} a_i a_j \mathbb{E}[U_i U_j]. \end{aligned}$$

Now $\mathbb{E}[U_i^2] = \text{Var}(Z_i)$, while the expected cross product is the covariance:

$$\mathbb{E}[U_i U_j] = \mathbb{E}[(Z_i - \mathbb{E}[Z_i])(Z_j - \mathbb{E}[Z_j])] = \text{Cov}(Z_i, Z_j).$$

Substituting these definitions completes the proof:

$$\text{Var} \left(\sum_i a_i Z_i \right) = \sum_i a_i^2 \text{Var}(Z_i) + 2 \sum_{i<j} a_i a_j \text{Cov}(Z_i, Z_j).$$

PART 3 / APPLYING THE VARIANCE FORMULA

When does averaging reduce uncertainty?

3.3 Apply the formula to the sample mean

Set every weight to $a_i = 1/n$ in the formula just proved. Then

$$\text{Var}(\hat{\mu}_n) = \frac{1}{n^2} \left[\sum_{i=1}^n \text{Var}(Z_i) + 2 \sum_{i < j} \text{Cov}(Z_i, Z_j) \right].$$

Why does independence eliminate the cross terms? If Z_i and Z_j are independent, their centered versions U_i and U_j are also independent. Since $\mathbb{E}[U_i] = \mathbb{E}[U_j] = 0$, the expectation of their product factors:

$$\text{Cov}(Z_i, Z_j) = \mathbb{E}[U_i U_j] = \mathbb{E}[U_i] \mathbb{E}[U_j] = 0.$$

Thus, for the independent losses in the question,

$$\text{Var}(\hat{\mu}_n) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(Z_i) \leq \frac{1}{n^2} \cdot \frac{n}{4} = \frac{1}{4n}.$$

Pairwise zero covariance is already sufficient for this variance simplification; dependence does *not* automatically imply nonzero covariance. Zero covariance alone does not justify the independent-variable Hoeffding bound. Expectation remains linear in all of these cases.

3.4 Two examples of dependence

Perfect positive dependence. Let $Z_i = B$ for every i , where B is Bernoulli with success probability $1/2$. Then

$$\hat{\mu}_n = B, \quad \text{Var}(\hat{\mu}_n) = \frac{1}{4}, \quad \mathbb{E}[D_n] = \frac{1}{2}, \quad \mathbb{P}(D_n \geq 0.1) = 1.$$

Each distinct pair has $\text{Cov}(Z_i, Z_j) = \text{Var}(B) = 1/4$. There are $n(n-1)/2$ pairs, so the formula confirms

$$\text{Var}(\hat{\mu}_n) = \frac{1}{n^2} \left[\frac{n}{4} + 2 \cdot \frac{n(n-1)}{2} \cdot \frac{1}{4} \right] = \frac{1}{4}.$$

The observations repeat the same random outcome, so increasing n gives no improvement.

Negative covariance can reduce variance. If $Z_1 = B$ and $Z_2 = 1 - B$, both means equal $1/2$. Their centered product is $(B - 1/2)(1/2 - B) = -(B - 1/2)^2$, so $\text{Cov}(Z_1, Z_2) = -1/4$. Consequently,

$$\text{Var}\left(\frac{Z_1 + Z_2}{2}\right) = \frac{1}{4} \left[\frac{1}{4} + \frac{1}{4} + 2 \left(-\frac{1}{4}\right) \right] = 0.$$

Indeed, their average is exactly $1/2$ for every outcome.

Markov, Jensen, and the identity $\mathbb{E}[D_n^2] = \text{Var}(\hat{\mu}_n)$ still apply under dependence when the common-mean assumption holds. What must be re-established is a variance bound that tends to zero. If $\text{Var}(\hat{\mu}_n) \rightarrow 0$, then $\mathbb{E}[D_n] \leq \sqrt{\text{Var}(\hat{\mu}_n)} \rightarrow 0$ even without independence.

References. The setting and numerical target are from *Week1_Question2_Question.pdf*, p. 1. For the convex and concave forms of Jensen, see Marco Taboga, “Jensen’s inequality,” StatLect (2021).